

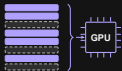
Naive Early Exit (No Speedup)

Batch of 8 Documents							
Doc 1				Doc 5			
Doc 2				Doc 6			
Doc 3				Doc 7			
Doc 4				Doc 8			

Transformer Layer (Exit Decision)

Tensor with Padding									
Doc 1					Doc 5				
Doc 2					0	0	Exited	0	0
0	0	Exited	0	0	Doc 7				
Doc 4					0	0	Exited	0	0

GPU Computation —
all 8 rows (3 wasted)



3 fewer rows
less GPU work

Triton Compaction (Real Speedup)

Batch of 8 Documents							
Doc 1				Doc 5			
Doc 2				Doc 6			
Doc 3				Doc 7			
Doc 4				Doc 8			

Transformer Layer (Exit Decision)

Compaction / Removal Step

Compact Tensor							
Doc 1							
Doc 2							
Doc 4							
Doc 5							
Doc 7							

GPU Computation —
only 5 rows (smaller tensor)

